

 Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Use appropriate smoothing and sharpening techniques to enhance the clarity of the images.	
OPED-9	Date: 31-08-2026	Enrollment No: 92400133030

GitHub Link:- <https://github.com/PG178/DSIP>

Code:-

```
import cv2

import numpy as np

import matplotlib.pyplot as plt

def apply_smoothing_filter(image, kernel_size):
    return cv2.blur(image, (kernel_size, kernel_size))

def apply_sharpening_filter(image):
    kernel = np.array([
        [0, -1, 0],
        [-1, 5, -1],
        [0, -1, 0]
    ])
    return cv2.filter2D(image, -1, kernel)

image_path = r"E:\DSIP\Oped-9\1.jpg"
input_image = cv2.imread(image_path)

if input_image is None:
    print("Error: 1.jpg not found!")
    exit()

smoothed_image = apply_smoothing_filter(
    input_image,
```

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```

kernel_size=5
)
sharpened_image =
    apply_sharpening_filter( input_image
)
input_rgb =
    cv2.cvtColor(
        input_image,
        cv2.COLOR_BGR2RGB
    )
smoothed_rgb =
    cv2.cvtColor(
        smoothed_image,
        cv2.COLOR_BGR2RGB
    )
sharpened_rgb =
    cv2.cvtColor(
        sharpened_image,
        cv2.COLOR_BGR2RGB
    )
plt.figure(figsize=(15, 5))
# Original
plt.subplot(1, 3, 1)
plt.imshow(input_rgb)

```

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plt.title("Original Image")

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```

plt.axis("off")
) #
Smoothed
plt.subplot(1, 3, 2)
plt.imshow(smoothed_rgb)
plt.title("Smoothed Image")
plt.axis("off")
# Sharpened
plt.subplot(1, 3, 3)
plt.imshow(sharpened_rgb)
plt.title("Sharpened Image")
plt.axis("off")
# Adjust spacing
plt.tight_layout()
) # Show output
plt.show()
cv2.imwrite(
    "smoothed_image.jpg"
    , smoothed_image
)
cv2.imwrite(
    "sharpened_image.jpg"
    , sharpened_image

```

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)

```
print("Smoothed image saved as: smoothed_image.jpg")
```

```
print("Sharpened image saved as: sharpened_image.jpg")
```

Output:-

Original Image



Smoothed Image



Sharpened Image



Original Image



Smoothed Image



Sharpened Image



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Original Image



Smoothed Image



Sharpened Image



Original Image



Smoothed Image



Sharpened Image



Original Image



Smoothed Image



Sharpened Image

